

Django Tutorial: Views & URLs Basics

Complete Handwritten-Style Master Notes | MVT Architecture, Views, URLs & include() Routing

Lecture #7

Hindi & Hinglish Guide

Beginner Friendly

Real-World Project Setup

1. Fundamental Concept: View & URL Kya Hote Hain?

Django **MVT (Model-View-Template)** architecture follow karta hai. Is video lecture ka main focus '**V**' (**View**) aur uske mapping partner **URL** par hai:

- **URL (Path / Address):** User browser mein jo address daalta hai (jaise `/blog/` ya `/about/`).
- **View (Brain / Logic):** Python ka ek function ya class jo request aane par run hota hai aur result (HTML ya text) return karta hai.



2. Real-World Project Setup (Terminal Commands)

Professional production environment ke hisaab se pehle virtual environment banana zaroori hota hai:

```
# 1. Virtual Environment create karna
python -m venv venv

# 2. Virtual Environment Activate karna
# Windows: venv\Scripts activate
# Mac/Linux: source venv/bin/activate

# 3. Django install karein
pip install django

# 4. Main Project banayein
django-admin startproject dj1
cd dj1

# 5. App create karein (jaise 'blog')
python manage.py startapp blog
```

Folder Structure Samjhein:

`dj1/` (Main Project directory) → Contains `manage.py`, `dj1/settings.py`, `dj1/urls.py`
`blog/` (Your App directory) → Contains `views.py`, `models.py`, `admin.py`

3. View Banane Ka Tareeqa (`blog/views.py`)

Har view function **request** accept karta hai aur hamesha **HttpResponse** return karta hai:

```

from django.http import HttpResponse

# Function 1: Simple string / HTML display
def home(request):
    return HttpResponse("<h1>Welcome to the Blog Home Page!</h1>")

# Function 2: Logic & Calculation execute karna
def about(request):
    a = 10 + 50
    return HttpResponse(f"About page output: {a}")

```

Golden Rule: View function ka pehla argument hamesha request hota hai! Bina request argument ke function error fekega.

4. URLs Ko Connect Karne Ke 2 Tareeqe

A Method 1: Project-Level urls.py mein direct import (Basic Way)

Direct view ko main project ke `django1/urls.py` mein jodna:

```

from django.contrib import admin
from django.urls import path
from blog import views # blog app se views import

urlpatterns = [
    path('admin/', admin.site.urls),
    path('blog/', views.home, name='home'),
    path('about/', views.about, name='about'),
]

```

Problem: Yeh method chhote projects ke liye theek hai, lekin bade projects jahan 10-15 apps hoti hain, wahan main `urls.py` bohot messy aur unmanageable ban jata hai.

B Method 2: App-Level urls.py + include() (Industry Standard / Best Way)

Isme har app ka apna ek separate `urls.py` hota hai, aur main project sirf `include()` karta hai.

Step 1: `blog/urls.py` file banayein aur app ke views map karein:

```

from django.urls import path
from . import views # Dot (.) means isi app folder ke views

urlpatterns = [
    path('', views.home, name='home'), # Base path: /blog/
    path('about/', views.about, name='about'), # Path: /blog/about/
]

```

Step 2: Main project ke `django1/urls.py` mein app ke URLs ko include karein:

```
from django.contrib import admin
from django.urls import path, include

urlpatterns = [
    path('admin/', admin.site.urls),
    path('blog/', include('blog.urls')), # Delegating to blog app
]
```

5. App Register Karna & Testing

Apni nayi banayi hui app ko `dj1/settings.py` mein register karein:

```
# dj1/settings.py
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    # ... default apps ...
    'blog', # ← Yahan apni app register karein
]
```

Server start karein:

```
python manage.py runserver
```

Browser URL test karein:

- `http://127.0.0.1:8000/blog/` → Displays **Welcome to the Blog Home Page!**
- `http://127.0.0.1:8000/blog/about/` → Displays **About page output: 60**

6. 🧠 Fast Revision Table (Yaad Rakhne Ke Punch Points)

Component	File	Role / Responsibility	Key Syntax
View Function	<code>views.py</code>	Request process karta hai, logic run karta hai aur <code>HttpResponse</code> return karta hai.	<code>def home(request): return HttpResponse(...)</code>
App URLs	<code>app/urls.py</code>	App ke functions ko local path mapping provide karta hai.	<code>path('about/', views.about, name='about')</code>
Project URLs	<code>project/urls.py</code>	Central router jo traffic ko respective app ke paas bhejta hai.	<code>path('blog/', include('blog.urls'))</code>
App Config	<code>settings.py</code>	Django ko batata hai ki kaunsi applications installed hain.	<code>INSTALLED_APPS = [... 'blog']</code>